

Project Report

Software Installation Request Automation in ServiceNow

Using Service Catalog and Workflows

Field	Details
Project Name	Software Installation Request Automation in ServiceNow
Project Type	Individual Project
Technology	ServiceNow
Primary Areas	Service Catalog, Workflow/Automation, Business Rules, Catalog Tables, Notifications, Update Sets
Objective	Automate software installation request intake, approval, fulfillment, tracking and exception handling.

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1. INTRODUCTION

1.1 Project Overview

The Software Installation Request Automation in ServiceNow project automates the process used by employees to request software installation. The solution uses a structured Service Catalog item, linked workflow/automation, automated task creation and centralized request tracking. It also provides a defined exception path for software-license shortages.

The source project objective is to streamline request submission, fulfillment tracking and issue management while using Update Sets so the configuration can be transferred between environments.

1.2 Purpose

The purpose is to reduce manual coordination between employees and IT, improve approval and fulfillment visibility, standardize request information, and support timely service delivery.

2. IDEATION PHASE

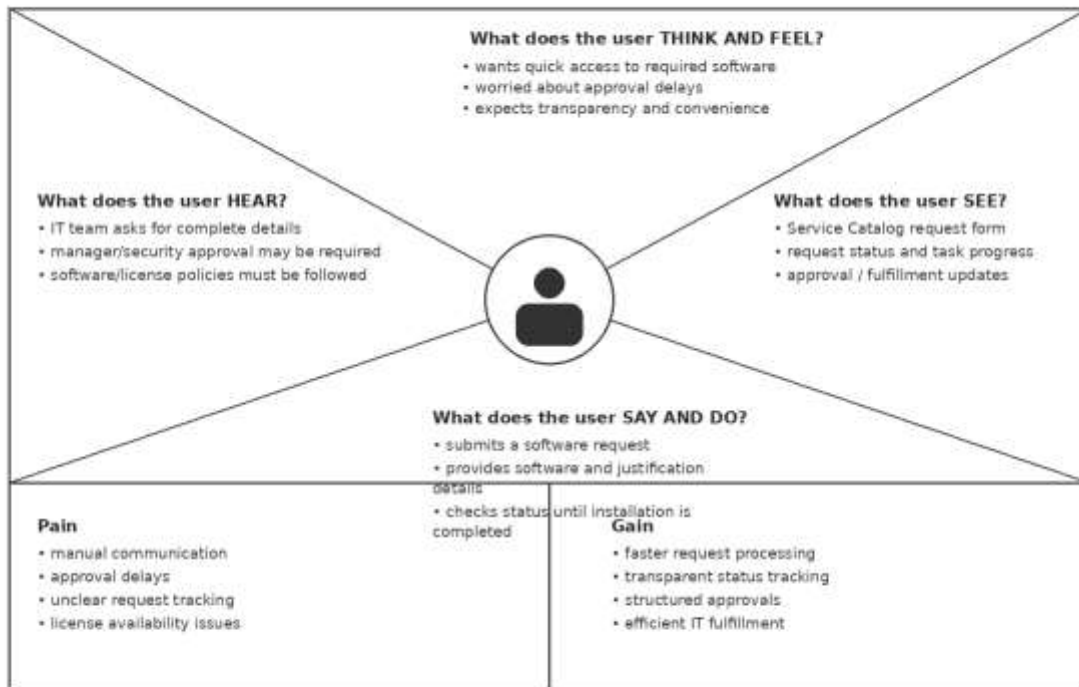
2.1 Problem Statement

Employees need a simple and transparent way to request software installation, but manual communication, approval delays, incomplete information, limited tracking, and license-related issues can slow fulfillment and increase IT workload.

User	I am	I'm trying to	But	Because	Which makes me feel
Employee / Requester	An employee who needs software for work	Request the required software installation through ServiceNow	Manual communication or approval delays may occur	There is no fully standardized and automated request path	Frustrated when the request takes longer or status is unclear
IT Support / Fulfillment	An IT support member handling software requests	Process and complete installation requests efficiently	Manual coordination and license issues can interrupt work	Request, task and exception handling require automation	Overloaded because coordination and tracking consume time

2.2 Empathy Map Canvas

The empathy map focuses on the employee requester and captures what the user thinks and feels, hears, sees and does, together with pain points and gains.



2.3 Brainstorming

Idea Group	Ideas
Request Intake	Service Catalog item; structured variables; simple request form
Approval & Governance	Approval workflow; security/compliance review
Fulfillment	Automatic catalog task; IT assignment; task-state tracking
Communication & Monitoring	Notifications; SLA monitoring; status visibility
Exceptions & Deployment	License-shortage incident; Update Sets
Reporting & Tracking	Lifecycle tracking; operational reporting

The highest-priority ideas are the Service Catalog item, structured variables, approval automation and catalog-task automation. Supporting ideas include notifications, SLA tracking, license exception handling and Update Sets.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Customer Journey Map - Software Installation Request



Key Insight

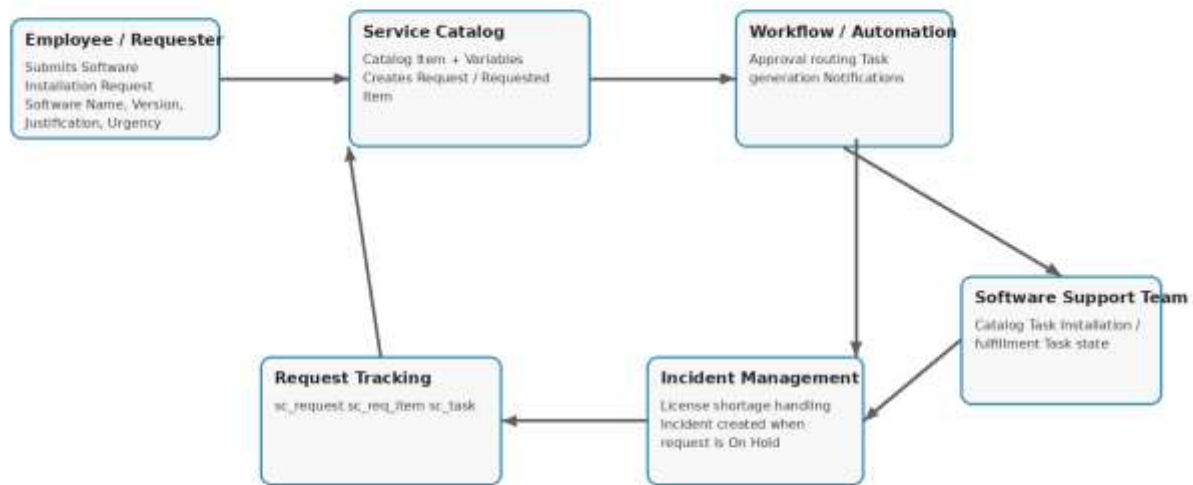
The solution should standardize request intake, automate approvals and fulfillment, improve visibility, and handle license issues consistently.

3.2 Solution Requirement

Requirement	Description
Catalog Item	Create a Service Catalog item for Software Installation Request.
Request Form	Capture Software Name, Version Required, License Justification and Urgency.
Approval	Route requests through the configured approval process.
Fulfillment Task	Create a catalog task for the Software Support / Fulfillment Team.
Tracking	Use request, requested-item and catalog-task records to track progress.
Notifications	Keep requesters, approvers and IT staff informed.
SLA Monitoring	Track timely fulfillment of software installation requests.
License Exception	When a request is placed On Hold due to license shortage, create an incident.
Deployment	Use Update Sets to package and transfer configuration.

3.3 Data Flow Diagram

Data Flow Diagram - Software Installation Request Automation



Flow: Employee -> Catalog Request -> Workflow/Approval -> Catalog Task -> Fulfillment / Exception Handling -> Completion & Tracking

3.4 Technology Stack

Component	Usage
ServiceNow Platform	Primary development and execution platform
Service Catalog	Request intake
Catalog Variables	Capture software request information
Workflow / Automation	Approval, task and lifecycle automation
sc_request / sc_req_item / sc_task	Request and fulfillment tracking
Business Rules	License shortage exception handling
Notifications	Lifecycle communication
Update Sets	Configuration packaging and transfer

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem - Solution Fit



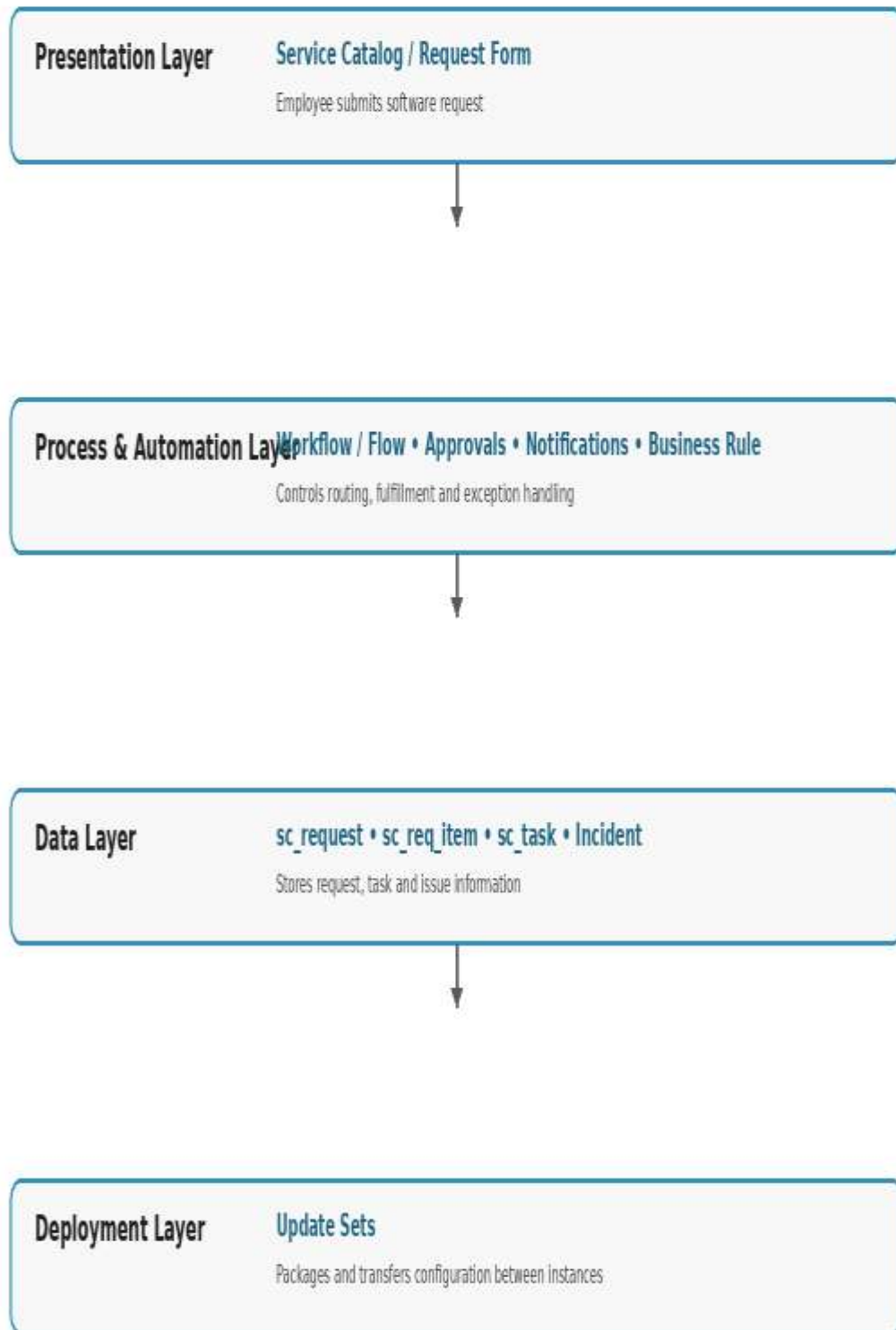
4.2 Proposed Solution

The proposed solution is a ServiceNow Service Catalog process for software installation. The employee submits a standardized form. The configured workflow/automation handles the approval path and fulfillment actions. A catalog task is assigned to the Software Support/Fulfillment Team. Requesters and IT staff can monitor the lifecycle using ServiceNow request and task records. If license availability prevents completion and the item is placed On Hold, a Business Rule creates an incident. Update Sets support movement of the configuration to another instance.

Stage	Configuration
Request	Software Installation Request catalog item
Input	Software Name; Version Required; License Justification; Urgency
Approval	Configured approval/workflow step
Fulfillment	Catalog task assigned to Software Support
Tracking	sc_request; sc_req_item; sc_task
Exception	Business Rule for On Hold / license shortage
Deployment	Complete -> Export XML -> Retrieve -> Preview -> Commit

4.3 Solution Architecture

Solution Architecture - ServiceNow



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project is planned as an individual two-Sprint implementation. Sprint 1 covers setup and core request automation; Sprint 2 covers validation, exception handling, testing and deployment.

Sprint	Epic	User Story / Task	Points	Priority
1	ServiceNow Setup & Catalog	Create Update Set	1	High
1	ServiceNow Setup & Catalog	Create Catalog Item	2	High
1	ServiceNow Setup & Catalog	Create Catalog Variables	3	High
1	Workflow & Fulfillment	Create Workflow / Flow	3	High
1	Workflow & Fulfillment	Attach Workflow to Catalog Item	2	High
1	Workflow & Fulfillment	Configure Catalog Task	3	High
2	Tables & Business Rules	Test sc_request and sc_req_item	2	High
2	Tables & Business Rules	Test sc_task assignment/state	2	High
2	Tables & Business Rules	Create License Shortage Business Rule	5	High
2	Testing & Deployment	Final Testing	3	High
2	Testing & Deployment	Complete and Export Update Set	2	Medium
2	Testing & Deployment	Retrieve, Preview and Commit Update Set	3	High

Sprint 1 Story Points = 14

Sprint 2 Story Points = 17

Total Story Points = 31

Number of Sprints = 2

Velocity = $31 / 2 = 15.5$ Story Points per Sprint

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing validates the functional execution and repeatability of the ServiceNow automation. This project does not use an AI/ML model, Power BI/DAX layer or Tableau workbook, so model-specific accuracy metrics are not applicable.

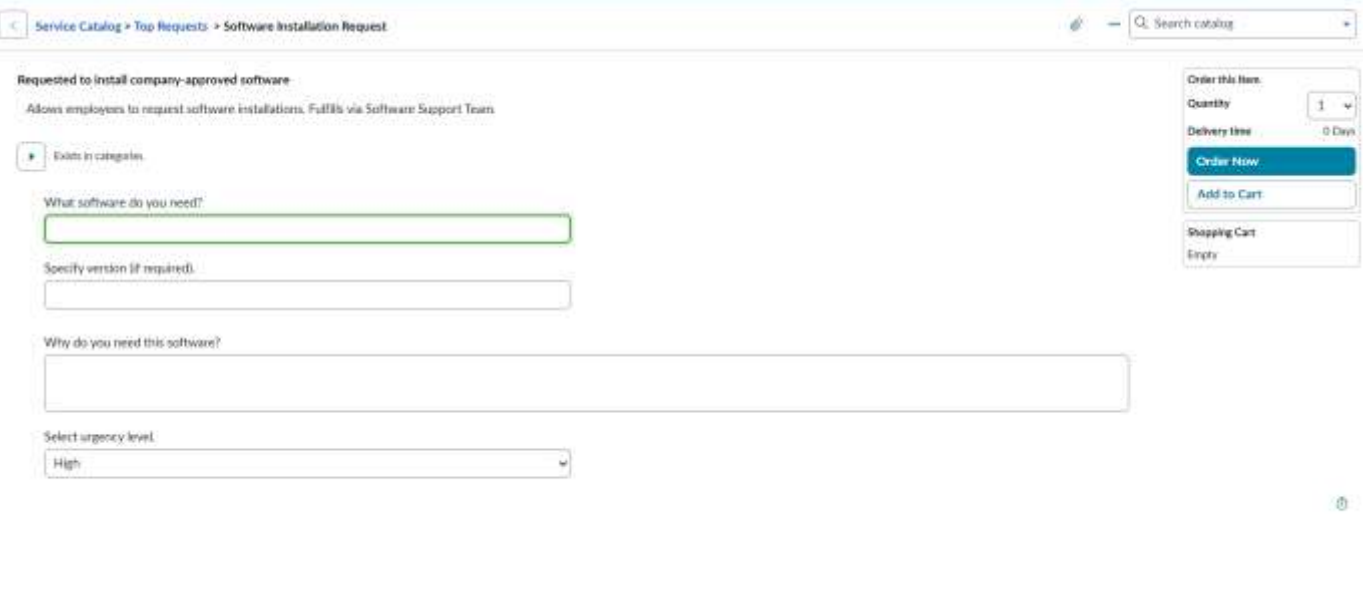
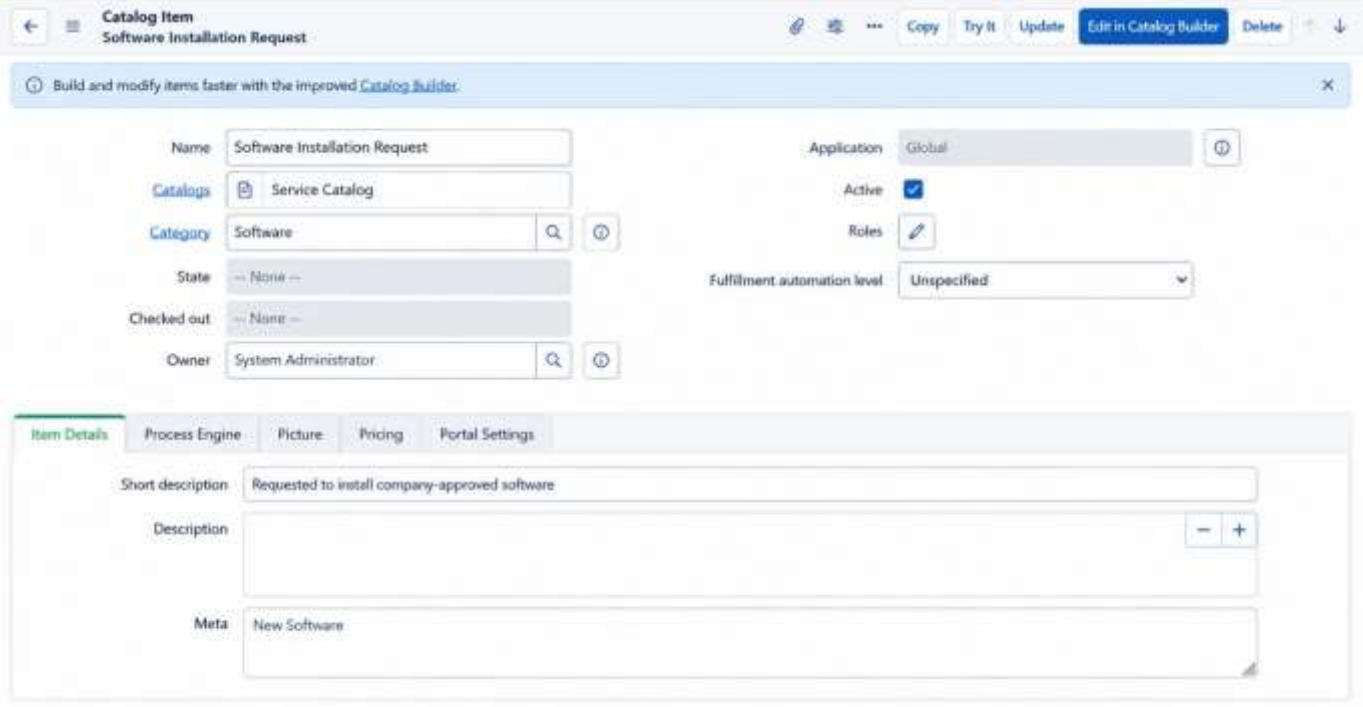
Test ID	Test Area	Test Action	Expected Result
FT-01	Catalog Item	Open Software Installation Request	Catalog item and expected variables are displayed
FT-02	Variables	Enter Software Name, Version, Justification and Urgency	Values are accepted and request can be submitted
FT-03	Request Records	Open the created request and requested item	sc_request and sc_req_item show the request information
FT-04	Approval / Workflow	Submit a request and observe automation	Configured approval/automation executes
FT-05	Catalog Task	Open the fulfillment task	Software Support assignment and expected description are present
FT-06	Task State	Update task state	State is saved and visible
FT-07	License Exception	Set requested item to On Hold	Incident is created by the Business Rule
FT-08	Update Set	Complete and export the Update Set	XML export is generated
FT-09	Update Set Import	Retrieve, preview and commit	Configuration transfers successfully

Performance validation is operational: repeated representative requests should execute consistently without record loss, task loss, or automation failure.

7. RESULTS

7.1 Output Screenshots

The supplied project document describes the required tests and expected records, but it does not include actual screenshots. Therefore, this section lists the exact evidence screenshots to capture from the completed ServiceNow instance rather than inventing results.

Screenshot	Evidence to Capture
1 	Software Installation Request catalog item and form variables
2 	Catalog item Try It / entered request data

3

Request

REQ0010003

Discuss

Follow

Update

Cancel Request

Copy

Delete

Number

REQ0010003

Opened

2026-09-04 06:38:36

Requested for

System Administrator

Opened by

System Administrator

Location

Approval

Approved

Due date

2026-09-04 06:38:36

Request state

Approved

Price

\$0.00

Description

Short description

Special instructions

Comments (Customer visible)

Comments (Customer visible)

Work notes

Post

Activities: 1

Field changes

2026-09-04 06:38:36

System Administrator

Impact

3 - Low

Opened by

System Administrator

Priority

4 - Low

Service Portal search and request form

4

Requests

Number

Search

Actions on selected rows...

New

All > Active = true

☐	Number	Requested for	Opened by	Request state	Due date
	Search	Search	Search	Search	Search
☐	REQ0010003	System Administrator	System Administrator	Approved	2026-09-04 06:38:36
☐	REQ0010002	System Administrator	System Administrator	Approved	2026-08-27 02:37:50
☐	REQ0010001	System Administrator	System Administrator	Approved	2026-08-27 02:34:29
☐	REQ0000001	System Administrator	System Administrator	Approved	2026-06-10 13:58:43

1

to 4 of 4

Generated Request / REQ number

5

Catalog Item
Software Installation Request

Copy Try It Update Edit in Catalog Builder Delete

Copy Try It Update Edit in Catalog Builder Delete

Related Links

- [Item Diagnostic](#)
- [Show VA render type](#)
- [Run Point Scan](#)

Variables (4) Variable Sets Catalog UI Policies Catalog Client Scripts Available For Not Available For Categories (1) Catalogs (1) Catalog Data Lookup Definitions Related Articles Related Catalog Items

Assigned Topics

Order Search Actions on selected rows... New

Catalog item = Software installation request

Type	Question	Order ▲
Single Line Text	What software do you need?	100
Single Line Text	Specify version (if required).	200
Multi Line Text	Why do you need this software?	300
Select Box	Select urgency level.	400

1 to 4 of 4

Requested
Item with
populated
catalog
variables

6

now

Workflow Studio

Software Request Workflow

Flow + Global

✕

+

Software Request Workflow

Active

🔒

View:

📄

🔗

Test

Edit flow

Deactivate

⋮

Action Properties

Action

Create Catalog Task

▼

Action Inputs

Table Name

Catalog Task [sc_task]

🔍

Requested Item [Requested Item]

Trigger... + Requested Item...

✕

📁

🔍

Short Description

📁

🔍

Fields

Assignment group

▼

Software

▼

⌚

🏠

📁

+

Wait

☒

📁

🔍

Template Catalog Item [Catalog Item]

Select Template Catalog Item

▼

🔍

Catalog Variables

Available

Select Template Catalog item field to load variables.

Selected

Create Catalog Item(s) that point to this flow and the variables on that item(s) will show

Data

Collapse All

Flow Variables

Trigger - Service Catalog

Requested Item Record

Record

Run Start Time UTC

Date/Time

Table Name

Table Name

Run Start Date/Time

Date/Time

1 - Ask For Approval

Approval State

Choice

Action Status

Object

2 - Create Catalog Task

Catalog Task

Record

Action Status

Object

Read-only

Status: Published

Application: Global

Catalog
Task
showing
Software
Support
assignment,
description
and state

7

Business Rule

Create Incident on License Issue

Update

Delete

When to run

Actions

Advanced

Specify whether the business rule should run on **Insert** or **Update**. Use **Filter Conditions** to specify under which conditions the business rule should run.

When

after

Order

100

Insert

☒

Update

☒

Delete

☐

Query

☐

Add another condition that must be true

Filter Conditions

Add Filter Condition

Add OR Clause

state

changes to

OnHold

AND

OR

X

Role conditions

Update

Delete

Related Links

[Run Point Scan](#)

Versions

Recorded at

Search

Actions on selected rows...

New

Update Versions

License-shortage test showing the generated incident

8

Update Set

Software Installation Request Setup

Update

Back Out

Name

Software Installation Request Setup

Application

Global

State

Complete

Created

2026-08-25 03:31:27

Parent

Created by

admin

Release date

Merged to

Install date

2026-08-25 03:31:21

Installed from

Description

Update

Back Out

Related Links

[Export to XML](#)
[Merge With Another Update Set](#)
[Scan Update Set](#)
[Show Update History](#)

Customer Updates (10)

Update Set Logs (19)

Child Update Sets

Install History

Created

Search

Actions on selected rows...

Update set = Software Installation Request Setup

	Created	Type	View	Target name	Updated by	Remote update set	Action
--	---------	------	------	-------------	------------	-------------------	--------

Completed Update Set and XML export

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9

<

Import XML

Importing records from an XML file will not run Business Rules.

Step 1: Choose file to upload

* XML file

Choose File

sys_remote_u....4013117.xml

Step 2: Upload the file

Upload

Retrieved Update Set in target instance

10

Retrieved Update Sets

Name

Search

Actions on selected rows...

All > Class = Retrieved Update Set

☐	Name	Application	State	Update source	Description	Loaded	Committed	Parent	Remote Batch Base
☐	Software Installation Request Setup	Global	Committed	(empty)		2026-08-25 08:26:20	2026-08-25 03:21:27	(empty)	(empty)

Related Links

[Import Update Set from XML](#)

1 to 1 of 1

Preview and Commit result with transferred configuration

Expected overall result: a standardized software installation request process with automated processing, transparent tracking, and structured license-issue handling.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Limitations
Standardized Service Catalog request intake	Initial configuration effort is required
Reduced manual coordination	Automation depends on correct configuration
Centralized request and task tracking	License availability remains an external constraint
Controlled approval and audit trail	Business Rules and workflow conditions require maintenance
Improved IT fulfillment visibility	Complex approval policies may require later enhancement
Structured license shortage handling	Accurate request data is required
Update Set-based transfer	Target instance still requires validation

9. CONCLUSION

The Software Installation Request Automation in ServiceNow project provides a structured and automated software-request lifecycle. Employees can submit software requests through the Service Catalog, while workflow/automation supports approvals and fulfillment. Catalog records and tasks provide centralized tracking, notifications improve communication, and the Business Rule provides a defined response to license shortages. Update Sets provide a practical way to transfer the configuration between instances.

The solution improves transparency, reduces manual coordination and provides a repeatable approach for handling software installation requests.

10. FUTURE SCOPE

- Introduce approval routing based on department, software type, cost, risk or requester role.
- Integrate license or asset information to validate availability before fulfillment.
- Add reminders and escalations for delayed approvals or fulfillment tasks.
- Add management dashboards for request volume, fulfillment time, SLA compliance and license issues.
- Integrate CMDB or endpoint information to validate the target device.
- Add stronger policy validation for restricted or high-risk software.
- Extend the catalog to additional software and access-request categories.

11. APPENDIX

Source Code(if any)

Business Rule code documented in the supplied project source:

```
if (current.state == 3) { // On Hold
    var inc = new GlideRecord('incident');
    inc.initialize();
    inc.short_description = 'License unavailable for ' + current.request_item;
    inc.description = 'Software request could not be fulfilled due to license
shortage.';
    inc.caller_id = current.request.requested_for;
    inc.assignment_group.setDisplayValue('Software Support');
    inc.priority = 3;
    inc.insert();
}
```

Dataset Link

Not applicable. This is a ServiceNow configuration and workflow automation project, not a dataset-based machine-learning project.

GitHub & Project Demo Link

<https://github.com/mercytalamala-171>

<https://drive.google.com/drive/folders/1YcOnt82DBIIjuf2Is2MdWS1zLa7SXDHA?usp=sharing>